Type MS · Finding the first term and common ratio of a geometric series

Answer 1 of 17 · Question 4 part a.ii · 5 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

4 (a) (i)	$\frac{ar^6}{ar^2}=\frac{40}{10}$	M1
	$r=\sqrt[4]{\frac{40}{10}}=(\pm\sqrt{2})$	M1
	$r=-\sqrt{2}$	A1
(ii)	$ar^2=10\Rightarrow a=\frac{10}{'2'}$	M1
	$a=5$	A1 (5)
(b)	$\lvert r\rvert>1$ so the series is not convergent.	B1FT (1)
		Total 6 marks

Part	Mark	Additional Guidance
(a)(i)	M1	For $\frac{ar^6}{ar^2}=\frac{40}{10}$ or $\frac{ar^2}{ar^6}=\frac{40}{10}$
	M1	For rearranging to find a value for r $r=\sqrt[4]{\frac{40}{10}}=(\pm\sqrt{2})$ or $\left(\pm\sqrt[4]{4}\right)$
	A1	For the correct value of $r=-\sqrt{2}$ Accept $-\sqrt[4]{4}$
(a)(ii)	M1	For attempting to find a value for a using their r $ar^2=10\Rightarrow a=\frac{10}{'2'}$
	A1	For the correct value of $a=5$ Allow this mark from the use of $r=\sqrt{2}$
(b)	B1FT	$\lvert r\rvert>1$ so the series is not convergent. Follow through their value of r provided it is greater than 1.

Type MS · Finding the first term and common ratio of a geometric series

Answer 1 of 17 · Question 4 part a.ii · 5 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 2 of 17 · Question 3 part a.i · 0 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part a.i in this mark scheme extract

3(a)(i)	$\frac{ar^4}{ar} = \frac{\frac{135}{1024}}{\frac{5}{16}}$ $r = \sqrt[3]{\frac{\frac{135}{1024}}{\frac{5}{16}}} = \left(\sqrt[3]{\frac{135\times16}{5\times1024}}\right) = \ldots$ $r = \frac{3}{4}$ oe	M1 M1 A1
(a)(ii)	$ar = \frac{5}{16} \Rightarrow a = \frac{\frac{5}{16}}{\frac{3}{4}} = \left(\frac{5}{12}\right)$ $a = \frac{5}{12}$	M1 A1 [5]
(b)	$S = \frac{\frac{5}{12}}{1-\frac{3}{4}} = \ldots$ $S = \frac{5}{3}$	M1 A1 [2]
		Total 7 marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 2 of 17 · Question 3 part a.i · 0 marks

2022 JAN · Paper 01 · extract 2 of 3 · find part a.i in this mark scheme extract

(b)	**M1**	For using the correct formula for the sum to infinity using their a and r provided $\|r\| < 1$ $S = \frac{'\frac{5}{12}'}{1-'\frac{3}{4}'} = \ldots$ $\frac{'5'}{12} \neq \frac{5}{16}$, $\frac{'5'}{12} \neq \frac{135}{1024}$
	A1	For the correct value of $S = \frac{5}{3}$ **Note:** Must be the exact value.

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Finding the first term and common ratio of a geometric series

Answer 2 of 17 · Question 3 part a.i · 0 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part a.i in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 3 of 17 · Question 3 part a.ii · 5 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part a.ii in this mark scheme extract

3(a)(i)	$\frac{ar^4}{ar} = \frac{\frac{135}{1024}}{\frac{5}{16}}$ $r = \sqrt[3]{\frac{\frac{135}{1024}}{\frac{5}{16}}} = \left(\sqrt[3]{\frac{135\times16}{5\times1024}}\right) = \ldots$ $r = \frac{3}{4}$ oe	M1 M1 A1
(a)(ii)	$ar = \frac{5}{16} \Rightarrow a = \frac{\frac{5}{16}}{,\frac{3}{4},} = \left(\frac{5}{12}\right)$ $a = \frac{5}{12}$	M1 A1 [5]
(b)	$S = \frac{,\frac{5}{12},}{1-,\frac{3}{4},} = \ldots$ $S = \frac{5}{3}$	M1 A1 [2]
		Total 7 marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 3 of 17 · Question 3 part a.ii · 5 marks

2022 JAN · Paper 01 · extract 2 of 3 · find part a.ii in this mark scheme extract

(b)	**M1**	For using the correct formula for the sum to infinity using their a and r provided $\|r\| < 1$ $S = \frac{'\frac{5}{12}'}{1-'\frac{3}{4}'} = \ldots$ $\frac{'5'}{12} \neq \frac{5}{16}$, $\frac{'5'}{12} \neq \frac{135}{1024}$
	A1	For the correct value of $S = \frac{5}{3}$ **Note:** Must be the exact value.

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Type MS · Finding the first term and common ratio of a geometric series

Answer 3 of 17 · Question 3 part a.ii · 5 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part a.ii in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 4 of 17 · Question 7 part a.i · 0 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a.i in this mark scheme extract

7 (a)	(i) $\frac{U_7}{U_3}=\frac{ar^6}{ar^2}=r^4\Rightarrow\frac{16}{4}=r^4\Rightarrow 4=r^4\Rightarrow 2=r^2\Rightarrow r=\sqrt{2}*$ (ii) $ar^2=4\Rightarrow a=\frac{4}{\left(\sqrt{2}\right)^2}=2$	M1 A1cso A1 [3]
Alternative		
(a)	(i) $ar^2=4\ ,\ ar^6=16\Rightarrow a\left(\frac{4}{a}\right)^3=16\Rightarrow a^2=4\Rightarrow a=2$ $r^2=\frac{4}{2}\Rightarrow r=\sqrt{2}*$ (ii) $a=2$	M1 A1cso A1
(b)	$500<2\times\sqrt{2}^{(n-1)}\Rightarrow 250<\sqrt{2}^{(n-1)}$ or $500<2^{\frac{1}{2}(n+1)}$ $\lg 250<(n-1)\lg\sqrt{2}$ or $\lg 500<\frac{1}{2}(n+1)\lg 2$ $n-1>\frac{\lg 250}{\lg\sqrt{2}}\Rightarrow n-1>15.93...$ or $\frac{1}{2}(n+1)>\frac{\lg 500}{\lg 2}\Rightarrow\frac{1}{2}(n+1)>8.96...$ $n>16.93...\Rightarrow n=17$	M1 M1 M1A1 [4]
(c)	$S_{20}=\frac{2\left(\left(\sqrt{2}\right)^{20}-1\right)}{\sqrt{2}-1}=\frac{2046}{\sqrt{2}-1}$ $\frac{2046}{\left(\sqrt{2}-1\right)}\times\frac{\left(\sqrt{2}+1\right)}{\left(\sqrt{2}+1\right)}=2046\sqrt{2}+2046=2046\left(1+\sqrt{2}\right)$	M1A1 M1A1 [4]
		Total 11 marks

(a)(i)	M1	$ar^2=4$ and $ar^6=16$
	A1 cso	Correct working to eliminate a and show that $r=\sqrt{2}$ (Answer given) Must see $r^4=4$.
(ii)	A1	$a=2$ (NB: This is not a B mark; must have scored the first M mark.)
Alternative		
(a)(i)	M1	$ar^2=4$ and $ar^6=16$
	A1 cso	Correct working to find a and use it to show that $r=\sqrt{2}$ (Answer given) Must see a correct substitution to eliminate r which is processed correctly to give $a=2$, and $r^2=\frac{4}{2}$ or $r^6=\frac{16}{2}$.
(ii)	A1	$a=2$ (Must have scored the first M mark.)

Type MS · Finding the first term and common ratio of a geometric series

Answer 4 of 17 · Question 7 part a.i · 0 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a.i in this mark scheme extract

(b)	M1	$500 < 2\times\left(\sqrt{2}\right)^{(n-1)}$ Accept an equation or >. Accept $500 < 2\times\sqrt{2}^{n-1}$ ft their value for a
	M1	$n-1 > \frac{\lg 250}{\lg\sqrt{2}}$ or $\frac{1}{2}(n-1) > \frac{\lg 250}{\lg 2}$ or $\frac{1}{2}(n+1) > \frac{\lg 500}{\lg 2}$ or $\lg_{\sqrt{2}} 250 < n-1$ Accept equation or >, and logs with any base. ft $500 < 2\times\left(\sqrt{2}\right)^{n}$
	M1	Evaluate and divide logs and attempt to find a value (at least 1DP) or numerical expression for n. Ignore incorrect inequality sign. eg $n > 16.9$ or $n = 16.9$ or $n > 1+15.93$ or $n = 1+2\times 7.96$ ft $500 < 2\times\left(\sqrt{2}\right)^{n}$
	A1	$n = 17$ Do not award after incorrect inequalities e.g. $n < 16.9$
Alternative		
(b)	M1	$500 < 2\times\left(\sqrt{2}\right)^{(n-1)}$ Accept an equation or >. Accept $500 < 2\times\sqrt{2}^{n-1}$ ft their value for a
	M1	Attempt to use trial and improvement, showing two trials that can be used to confirm the required value of n, eg $\left(\sqrt{2}\right)^{15} = 181.0...$ and $\left(\sqrt{2}\right)^{16} = 256$ or $2\left(\sqrt{2}\right)^{15} = 362.0...$ and $2\left(\sqrt{2}\right)^{16} = 512$ or $2^8 = 256$ and $2^9 = 512$ ft $500 < 2\times\left(\sqrt{2}\right)^{n}$ if clearly stated
	M1	Use the higher power from these trials in an appropriate equation eg $n-1 = 16$ or $\frac{1}{2}(n+1) = 9$ and attempt to find a value for n. ft $500 < 2\times\left(\sqrt{2}\right)^{n}$ if clearly stated
	A1	$n = 17$
(c)	M1	Attempt to use $S_n = \frac{a(1-r^n)}{(1-r)}$ with their a, $r = \sqrt{2}$ and $n = 20$ Allow notation or sign errors if formula is stated, otherwise the substitution must be correct.
	A1	$S_{20} = \frac{2046}{\sqrt{2}-1}$ or $S_{20} = -\frac{2046}{1-\sqrt{2}}$ Correct expression with numerator evaluated.
	M1	Attempt to rationalise the denominator. Must use $\sqrt{2}+1$ and attempt to simplify the denominator.
	A1	$2046(1+\sqrt{2})$ or $2046(\sqrt{2}+1)$ Accept $p = 2046$. Award full marks for a correct answer if first M1 has been awarded.

Type MS · Finding the first term and common ratio of a geometric series

Answer 4 of 17 · Question 7 part a.i · 0 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 5 of 17 · Question 7 part a.ii · 3 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a.ii in this mark scheme extract

7 (a)	(i) $\frac{U_7}{U_3}=\frac{ar^6}{ar^2}=r^4\Rightarrow\frac{16}{4}=r^4\Rightarrow 4=r^4\Rightarrow 2=r^2\Rightarrow r=\sqrt{2}*$ (ii) $ar^2=4\Rightarrow a=\frac{4}{\left(\sqrt{2}\right)^2}=2$	M1 A1cso A1 [3]
Alternative		
(a)	(i) $ar^2=4\ ,\ ar^6=16\Rightarrow a\left(\frac{4}{a}\right)^3=16\Rightarrow a^2=4\Rightarrow a=2$ $r^2=\frac{4}{2}\Rightarrow r=\sqrt{2}*$ (ii) $a=2$	M1 A1cso A1
(b)	$500<2\times\sqrt{2}^{(n-1)}\Rightarrow 250<\sqrt{2}^{(n-1)}$ or $500<2^{\frac{1}{2}(n+1)}$ $\lg 250<(n-1)\lg\sqrt{2}$ or $\lg 500<\frac{1}{2}(n+1)\lg 2$ $n-1>\frac{\lg 250}{\lg\sqrt{2}}\Rightarrow n-1>15.93...$ or $\frac{1}{2}(n+1)>\frac{\lg 500}{\lg 2}\Rightarrow\frac{1}{2}(n+1)>8.96...$ $n>16.93...\Rightarrow n=17$	M1 M1 M1A1 [4]
(c)	$S_{20}=\frac{2\left(\left(\sqrt{2}\right)^{20}-1\right)}{\sqrt{2}-1}=\frac{2046}{\sqrt{2}-1}$ $\frac{2046}{\left(\sqrt{2}-1\right)}\times\frac{\left(\sqrt{2}+1\right)}{\left(\sqrt{2}+1\right)}=2046\sqrt{2}+2046=2046\left(1+\sqrt{2}\right)$	M1A1 M1A1 [4]
		Total 11 marks

(a)(i)	M1	$ar^2=4$ and $ar^6=16$
	A1 cso	Correct working to eliminate a and show that $r=\sqrt{2}$ (Answer given) Must see $r^4=4$.
(ii)	A1	$a=2$ (NB: This is not a B mark; must have scored the first M mark.)
Alternative		
(a)(i)	M1	$ar^2=4$ and $ar^6=16$
	A1 cso	Correct working to find a and use it to show that $r=\sqrt{2}$ (Answer given) Must see a correct substitution to eliminate r which is processed correctly to give $a=2$, and $r^2=\frac{4}{2}$ or $r^6=\frac{16}{2}$.
(ii)	A1	$a=2$ (Must have scored the first M mark.)

Type MS · Finding the first term and common ratio of a geometric series

Answer 5 of 17 · Question 7 part a.ii · 3 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a.ii in this mark scheme extract

(b)	M1	$500 < 2\times\left(\sqrt{2}\right)^{(n-1)}$ Accept an equation or >. Accept $500 < 2\times\sqrt{2}^{n-1}$ ft their value for a
	M1	$n-1 > \frac{\lg 250}{\lg\sqrt{2}}$ or $\frac{1}{2}(n-1) > \frac{\lg 250}{\lg 2}$ or $\frac{1}{2}(n+1) > \frac{\lg 500}{\lg 2}$ or $\lg_{\sqrt{2}} 250 < n-1$ Accept equation or >, and logs with any base. ft $500 < 2\times\left(\sqrt{2}\right)^{n}$
	M1	Evaluate and divide logs and attempt to find a value (at least 1DP) or numerical expression for n. Ignore incorrect inequality sign. eg $n > 16.9$ or $n = 16.9$ or $n > 1+15.93$ or $n = 1+2\times 7.96$ ft $500 < 2\times\left(\sqrt{2}\right)^{n}$
	A1	$n = 17$ Do not award after incorrect inequalities e.g. $n < 16.9$
Alternative		
(b)	M1	$500 < 2\times\left(\sqrt{2}\right)^{(n-1)}$ Accept an equation or >. Accept $500 < 2\times\sqrt{2}^{n-1}$ ft their value for a
	M1	Attempt to use trial and improvement, showing two trials that can be used to confirm the required value of n, eg $\left(\sqrt{2}\right)^{15} = 181.0...$ and $\left(\sqrt{2}\right)^{16} = 256$ or $2\left(\sqrt{2}\right)^{15} = 362.0...$ and $2\left(\sqrt{2}\right)^{16} = 512$ or $2^8 = 256$ and $2^9 = 512$ ft $500 < 2\times\left(\sqrt{2}\right)^{n}$ if clearly stated
	M1	Use the higher power from these trials in an appropriate equation eg $n-1 = 16$ or $\frac{1}{2}(n+1) = 9$ and attempt to find a value for n. ft $500 < 2\times\left(\sqrt{2}\right)^{n}$ if clearly stated
	A1	$n = 17$
(c)	M1	Attempt to use $S_n = \frac{a(1-r^n)}{(1-r)}$ with their a, $r = \sqrt{2}$ and $n = 20$ Allow notation or sign errors if formula is stated, otherwise the substitution must be correct.
	A1	$S_{20} = \frac{2046}{\sqrt{2}-1}$ or $S_{20} = -\frac{2046}{1-\sqrt{2}}$ Correct expression with numerator evaluated.
	M1	Attempt to rationalise the denominator. Must use $\sqrt{2}+1$ and attempt to simplify the denominator.
	A1	$2046(1+\sqrt{2})$ or $2046(\sqrt{2}+1)$ Accept $p = 2046$. Award full marks for a correct answer if first M1 has been awarded.

Type MS · Finding the first term and common ratio of a geometric series

Answer 5 of 17 · Question 7 part a.ii · 3 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 6 of 17 · Question 5 part b · 3 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

5 (a)	$[S_\infty =]\dfrac{12}{1-\frac{3}{8}} = \dfrac{96}{5}$	M1 A1 (2)
(b)	$ar^5 = 12\left(\dfrac{3}{8}\right)^5$	M1
	$= \dfrac{2^2\times 3\times 3^5}{2^{15}} = \dfrac{3^6}{2^{13}}$ *	M1 A1cso (3)
(c)	e.g. $u_n = 12\left(\dfrac{3}{8}\right)^{n-1}$	M1
	$\log_2 u_n = \log_2 12 + \log_2\left(\dfrac{3}{8}\right)^{n-1}$	M1
	$\log_2 u_n = \log_2 12 + (n-1)\left[\log_2 3 - \log_2 8\right]$	M1
	$\log_2 u_n = \log_2 3 + 2\log_2 2 + (n-1)[\log_2 3 - 3\log_2 2]$	M1
	$\log_2 u_n = n\log_2 3 - 3n + 5$ *	A1 cso (5)
		Total 10 marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 6 of 17 · Question 5 part b · 3 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For the correct use of $\frac{a}{1-r}$
	A1	For the correct value $\frac{96}{5}$
(b)	M1	For the correct use of ar^{n-1}
	M1	For expressing 12 $\left(2^2\times 3\right)$ and 8 $\left(2^3\right)$ as powers of 2
	A1cso	For obtaining the given expression with no errors seen.
(c)	M1	For the correct use of ar^{n-1} and the given values of a and r to give $u_n = 12\left(\frac{3}{8}\right)^{n-1}$
	M1	For taking logs [base 2] of both sides and applying the addition law.
	M1	For applying the power law and subtraction laws to $\log_2\left(\frac{3}{8}\right)^{n-1}$ $\log_2\left(\frac{3}{8}\right)^{n-1} = (n-1)\left[\log_2 3 - \log_2 8\right]$
	M1	For obtaining $\log_2 12 = \log_2 3 + 2\log_2 2$
	A1cso	For obtaining the given equation with no errors seen.
	ALT	
	M1	For the correct use of ar^{n-1} and the given values of a and r to give $u_n = 12\left(\frac{3}{8}\right)^{n-1}$
	M1	For rearranging the equation to obtain: $U_n = 12\left(\frac{3}{8}\right)^{n-1} = 2^2\times 3\times\frac{3^{n-1}}{2^{3(n-1)}} = \frac{3^n}{2^{3n-5}}$
	M1	For taking logs of both sides and applying the addition (subtraction) law. $\log_2 U_n = \log_2 3^n - \log_2\left(3n-5\right)$
	M1	For applying the power law to obtain: $\log_2 U_n = n\log_2 3 - \log_2\left(3n-5\right)$
	A1 cso	For obtaining the given equation with no errors seen.

Type MS · Finding the first term and common ratio of a geometric series

Answer 6 of 17 · Question 5 part b · 3 marks

2022 JAN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 7 of 17 · Question 4 part a.i · 0 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

4 (a) (i)	$\frac{ar^6}{ar^2}=\frac{40}{10}$	M1
	$r=\sqrt[4]{\frac{40}{10}}=(\pm\sqrt{2})$	M1
	$r=-\sqrt{2}$	A1
(ii)	$ar^2=10\Rightarrow a=\frac{10}{'2'}$	M1
	$a=5$	A1 (5)
(b)	$\lvert r\rvert>1$ so the series is not convergent.	B1FT (1)
		Total 6 marks

Part	Mark	Additional Guidance
(a)(i)	M1	For $\frac{ar^6}{ar^2}=\frac{40}{10}$ or $\frac{ar^2}{ar^6}=\frac{40}{10}$
	M1	For rearranging to find a value for r $r=\sqrt[4]{\frac{40}{10}}=(\pm\sqrt{2})$ or $\left(\pm\sqrt[4]{4}\right)$
	A1	For the correct value of $r=-\sqrt{2}$ Accept $-\sqrt[4]{4}$
(a)(ii)	M1	For attempting to find a value for a using their r $ar^2=10\Rightarrow a=\frac{10}{'2'}$
	A1	For the correct value of $a=5$ Allow this mark from the use of $r=\sqrt{2}$
(b)	B1FT	$\lvert r\rvert>1$ so the series is not convergent. Follow through their value of r provided it is greater than 1.

Type MS · Finding the first term and common ratio of a geometric series

Answer 7 of 17 · Question 4 part a.i · 0 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 8 of 17 · Question 7 part a · 3 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

7 (a)	$\frac{ar^6}{ar^3} = r^3 = \frac{e^{\frac{2x+1}{2}}}{e^{x+2}} = \frac{e^x \times e^{\frac{1}{2}}}{e^x \times e^2} = e^{-\frac{3}{2}} \Rightarrow r = e^{-\frac{1}{2}}$ * **ALT** $\frac{e^{x+2}}{r^3} r^6 = e^{\frac{2x+1}{2}} \Rightarrow r^3 = e^{x+\frac{1}{2}-x+2} \Rightarrow r^3 = e^{-\frac{3}{2}} \Rightarrow r = e^{-\frac{1}{2}}$ *	M1M1A1 cso [3] {M1} {M1}{A1} cso [3]
(b)	$ar^3 = ae^{-\frac{3}{2}} = e^{x+2} \Rightarrow a = \frac{e^{x+2}}{e^{-\frac{3}{2}}} = \frac{e^x \times e^2}{e^{-\frac{3}{2}}} = e^{x+\frac{7}{2}}$ oe **ALT** $a = \frac{e^{x+2}}{e^{-\frac{3}{2}}} = e^{x+2+\frac{3}{2}} = e^{x+\frac{7}{2}}$ oe	M1M1A1 [3] {M1}{M1} {A1} [3]

(a)	
M1	Use of $\frac{ar^6}{ar^3} = r^3$
M1	Using $e^{a+b} = e^a \times e^b$ to simplify leading to $r^3 = e^c$ where c is a number
A1 cso	Obtains the **given** answer with no errors in the working
ALT	
M1	For rearranging to make a subject and substituting into the other equation
M1	Using $e^{a-b} = e^a \div e^b$ to simplify leading to $r^3 = e^c$ where c is a number
A1 cso	Obtains the **given** answer with no errors in the working
(b)	
M1	For $a = \frac{e^{x+2}}{e^{-\frac{3}{2}}}$
M1	Using $e^{a+b} = e^a \times e^b$ to simplify to $a = \frac{e^x \times e^2}{e^{-\frac{3}{2}}}$
A1	$e^{x+\frac{7}{2}}$ oe
ALT	
M1	For $a = \frac{e^{x+2}}{e^{-\frac{3}{2}}}$
M1	Using $e^{a+b} = e^a \times e^b$ to simplify to $e^{x+2+\frac{3}{2}}$
A1	$e^{x+\frac{7}{2}}$ oe

Type MS · Finding the first term and common ratio of a geometric series

Answer 8 of 17 · Question 7 part a · 3 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

(c)	$S_\infty = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} = \dfrac{e^{x+\frac{7}{2}}}{\frac{e^{\frac{1}{2}}-1}{e^{\frac{1}{2}}}} = \dfrac{e^{x+4}}{e^{\frac{1}{2}}-1} \Rightarrow p = x+4$ **ALT 1** $S_\infty = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} \times \dfrac{e^{\frac{1}{2}}}{e^{\frac{1}{2}}} = \dfrac{e^{x+4}}{e^{\frac{1}{2}}-1} \Rightarrow p = x+4$ **ALT 2** $S_\infty = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} = \dfrac{e^{p}}{e^{\frac{1}{2}}-1} \Rightarrow e^{x+\frac{7}{2}}\left(e^{\frac{1}{2}}-1\right) = e^{p}\left(1-e^{-\frac{1}{2}}\right)$ $\Rightarrow e^{x+4} - e^{x+\frac{7}{2}} = e^{p} - e^{p-\frac{1}{2}} \Rightarrow p = x+4$	M1M1A1 [3] {M1}{M1} {A1} [3] {M1}{M1} {A1} [3]

(c)	
M1	Use of $S_\infty = \dfrac{a}{1-r}$
M1	$1-e^{-\frac{1}{2}} = \dfrac{e^{\frac{1}{2}}-1}{e^{\frac{1}{2}}}$
A1	$p = x+4$
ALT 1	
M1	Use of $S_\infty = \dfrac{a}{1-r}$
M1	Multiplying S_∞ by $\dfrac{e^{\frac{1}{2}}}{e^{\frac{1}{2}}}$
A1	$p = x+4$
ALT 2	
M1	Use of $S_\infty = \dfrac{a}{1-r}$
M1	For simplifying to $e^{x+4} - e^{x+\frac{7}{2}} = e^{p} - e^{p-\frac{1}{2}}$ oe
A1	$p = x+4$

Type MS · Finding the first term and common ratio of a geometric series

Answer 8 of 17 · Question 7 part a · 3 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

(d)	$e^{x+\frac{7}{2}} \times \left(e^{-\frac{1}{2}}\right)^{17} > 1.6 \Rightarrow e^x \times e^{\left(\frac{7}{2}-\frac{17}{2}\right)} > 1.6 \Rightarrow e^x > \frac{1.6}{e^{-5}}$	M1
	$\Rightarrow e^x > 237.46105....$	M1
	$\Rightarrow x > \ln(237.46105) \Rightarrow x > 5.4700...$	M1
	$\Rightarrow x = 6$	A1
		[4]
	ALT	
	$e^{x+\frac{7}{2}} \times \left(e^{-\frac{1}{2}}\right)^{17} > 1.6 \Rightarrow e^{x-5} > 1.6$	{M1}
	$\Rightarrow x - 5 > \ln(1.6)$	{M1}
	$\Rightarrow x > \ln(1.6) + 5 \Rightarrow x > 5.4700...$	{M1}
	$\Rightarrow x = 6$	{A1}
		[4]
		Total 13 marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 8 of 17 · Question 7 part a · 3 marks

2020 JAN · Paper 02 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 9 of 17 · Question 7 part b · 3 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

7 (a)	$\frac{ar^6}{ar^3} = r^3 = \frac{e^{\frac{2x+1}{2}}}{e^{x+2}} = \frac{e^x \times e^{\frac{1}{2}}}{e^x \times e^2} = e^{-\frac{3}{2}} \Rightarrow r = e^{-\frac{1}{2}}$ * **ALT** $\frac{e^{x+2}}{r^3} r^6 = e^{\frac{2x+1}{2}} \Rightarrow r^3 = e^{x+\frac{1}{2}-x+2} \Rightarrow r^3 = e^{-\frac{3}{2}} \Rightarrow r = e^{-\frac{1}{2}}$ *	M1M1A1 cso [3] {M1} {M1}{A1} cso [3]
(b)	$ar^3 = ae^{-\frac{3}{2}} = e^{x+2} \Rightarrow a = \frac{e^{x+2}}{e^{-\frac{3}{2}}} = \frac{e^x \times e^2}{e^{-\frac{3}{2}}} = e^{x+\frac{7}{2}}$ oe **ALT** $a = \frac{e^{x+2}}{e^{-\frac{3}{2}}} = e^{x+2+\frac{3}{2}} = e^{x+\frac{7}{2}}$ oe	M1M1A1 [3] {M1}{M1} {A1} [3]

(a)

M1 Use of $\frac{ar^6}{ar^3} = r^3$

M1 Using $e^{a+b} = e^a \times e^b$ to simplify leading to $r^3 = e^c$ where c is a number

A1 cso Obtains the **given** answer with no errors in the working

ALT

M1 For rearranging to make a subject and substituting into the other equation

M1 Using $e^{a-b} = e^a \div e^b$ to simplify leading to $r^3 = e^c$ where c is a number

A1 cso Obtains the **given** answer with no errors in the working

(b)

M1 For $a = \frac{e^{x+2}}{e^{-\frac{3}{2}}}$

M1 Using $e^{a+b} = e^a \times e^b$ to simplify to $a = \frac{e^x \times e^2}{e^{-\frac{3}{2}}}$

A1 $e^{x+\frac{7}{2}}$ oe

ALT

M1 For $a = \frac{e^{x+2}}{e^{-\frac{3}{2}}}$

M1 Using $e^{a+b} = e^a \times e^b$ to simplify to $e^{x+2+\frac{3}{2}}$

A1 $e^{x+\frac{7}{2}}$ oe

Type MS · Finding the first term and common ratio of a geometric series

Answer 9 of 17 · Question 7 part b · 3 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

(c)	$S_\infty = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} = \dfrac{e^{x+\frac{7}{2}}}{\frac{e^{\frac{1}{2}}-1}{e^{\frac{1}{2}}}} = \dfrac{e^{x+4}}{e^{\frac{1}{2}}-1} \Rightarrow p = x+4$ **ALT 1** $S_\infty = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} \times \dfrac{e^{\frac{1}{2}}}{e^{\frac{1}{2}}} = \dfrac{e^{x+4}}{e^{\frac{1}{2}}-1} \Rightarrow p = x+4$ **ALT 2** $S_\infty = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} = \dfrac{e^{p}}{e^{\frac{1}{2}}-1} \Rightarrow e^{x+\frac{7}{2}}\left(e^{\frac{1}{2}}-1\right) = e^{p}\left(1-e^{-\frac{1}{2}}\right)$ $\Rightarrow e^{x+4} - e^{x+\frac{7}{2}} = e^{p} - e^{p-\frac{1}{2}} \Rightarrow p = x+4$	M1M1A1 [3] {M1}{M1} {A1} [3] {M1}{M1} {A1} [3]
(c)		
M1	Use of $S_\infty = \dfrac{a}{1-r}$	
M1	$1-e^{-\frac{1}{2}} = \dfrac{e^{\frac{1}{2}}-1}{e^{\frac{1}{2}}}$	
A1	$p = x+4$	
ALT 1		
M1	Use of $S_\infty = \dfrac{a}{1-r}$	
M1	Multiplying S_∞ by $\dfrac{e^{\frac{1}{2}}}{e^{\frac{1}{2}}}$	
A1	$p = x+4$	
ALT 2		
M1	Use of $S_\infty = \dfrac{a}{1-r}$	
M1	For simplifying to $e^{x+4} - e^{x+\frac{7}{2}} = e^{p} - e^{p-\frac{1}{2}}$ oe	
A1	$p = x+4$	

Type MS · Finding the first term and common ratio of a geometric series

Answer 9 of 17 · Question 7 part b · 3 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

(d)	$e^{x+\frac{7}{2}} \times \left(e^{-\frac{1}{2}}\right)^{17} > 1.6 \Rightarrow e^x \times e^{\left(\frac{7}{2}-\frac{17}{2}\right)} > 1.6 \Rightarrow e^x > \frac{1.6}{e^{-5}}$	M1
	$\Rightarrow e^x > 237.46105....$	M1
	$\Rightarrow x > \ln(237.46105) \Rightarrow x > 5.4700...$	M1
	$\Rightarrow x = 6$	A1
		[4]
	ALT	
	$e^{x+\frac{7}{2}} \times \left(e^{-\frac{1}{2}}\right)^{17} > 1.6 \Rightarrow e^{x-5} > 1.6$	{M1}
	$\Rightarrow x - 5 > \ln(1.6)$	{M1}
	$\Rightarrow x > \ln(1.6) + 5 \Rightarrow x > 5.4700...$	{M1}
	$\Rightarrow x = 6$	{A1}
		[4]
		Total 13 marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 9 of 17 · Question 7 part b · 3 marks

2020 JAN · Paper 02 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 10 of 17 · Question 3 part a.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

3 (a) (i)	$ar^2 = 5$ or $ar^4 = \frac{5}{2}$ or $5r^2 = \frac{5}{2}$	B1
	$\left(\frac{ar^4}{ar^2}\right) = \frac{\frac{5}{2}}{5}$ or $\frac{5}{r^2} = \frac{\frac{5}{2}}{r^4}$ oe $\rightarrow r$ or $r = \sqrt{\frac{\frac{5}{2}}{5}}$	M1
	$r = \frac{\sqrt{2}}{2}$ oe	A1
(ii)	$a = 10$	A1 [4]
(b)	$S_\infty = \frac{"10"}{1-"\frac{\sqrt{2}}{2}"}$	M1
	$20 + 10\sqrt{2}$	A1 [2]
		Total 6 marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 10 of 17 · Question 3 part a.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 11 of 17 · Question 3 part a.ii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

3 (a) (i)	$ar^2 = 5$ or $ar^4 = \frac{5}{2}$ or $5r^2 = \frac{5}{2}$	B1
	$\left(\frac{ar^4}{ar^2}\right) = \frac{\frac{5}{2}}{5}$ or $\frac{5}{r^2} = \frac{\frac{5}{2}}{r^4}$ oe $\rightarrow r$ or $r = \sqrt{\frac{\frac{5}{2}}{5}}$	M1
	$r = \frac{\sqrt{2}}{2}$ oe	A1
(ii)	$a = 10$	A1 [4]
(b)	$S_\infty = \frac{"10"}{1-"\frac{\sqrt{2}}{2}"}$	M1
	$20 + 10\sqrt{2}$	A1 [2]
		Total 6 marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 11 of 17 · Question 3 part a.ii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 12 of 17 · Question 7 part a · 3 marks

2020 JAN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

7(a)	$\frac{ar^7}{ar^6} = \frac{1152}{192} \quad (= 6) = r$	B1
	4th term $= \frac{192}{6^3}$ or $\frac{1152}{6^4} = \frac{8}{9}$	M1A1 (3)
(b)	$\frac{t_3}{r} + t_3 + rt_3 \Rightarrow \frac{24}{r} + 24 + 24r = -36$	M1A1 NB B1B1 on e-PEN
	$24 + 24r + 24r^2 = -36r$	dM1
	$24r^2 + 60r + 24 = 2r^2 + 5r + 2 = 0$ *	ddM1A1cso (5)
(c)	$2r^2 + 5r + 2 = 0 \Rightarrow (2r + 1)(r + 2) = 0 \Rightarrow r = -\frac{1}{2}$	M1A1
	$S = \frac{a}{1-r} = \frac{24 \div \left(-\frac{1}{2}\right)^2}{1-\left(-\frac{1}{2}\right)}, = 64$	M1,A1 (4)
		[12]

(a)	
B1	Obtain a correct value for r. Fraction need not be simplified
M1	Use their r and either the 7th or 8th term divided by the appropriate power of r to obtain the 4th term as a fraction – no need to simplify
A1	$\frac{8}{9}$
ALT	M1 Find a (=1/243) and use ar^3 A1 Correct answer
(b)	
M1	Use the given information to obtain an equation in r
A1	Correct equation
dM1	Eliminate the fraction
ddM1	Obtain a 3TQ, terms in any order
A1cso	Reach the given result with no errors in the working
(c)	
M1	Solve the given quadratic by any valid method. Must reach a value of r
A1	Correct value of r (Ignore second answer if given)
M1	Use the formula for the sum to infinity with their r **provided** $\lvert r \rvert < 1$. a must be 24 divided by (their r)^2
A1	Correct answer.

Type MS · Finding the first term and common ratio of a geometric series

Answer 12 of 17 · Question 7 part a · 3 marks

2020 JAN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question Number	Scheme	Marks
	$\frac{dy}{}$	

Type MS · Finding the first term and common ratio of a geometric series

Answer 13 of 17 · Question 8 part a · 1 mark

2023 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

8(a)	$U_5 = \left(\frac{25}{4}\right)\left(\frac{3}{5}\right)^5 = \frac{243}{500}$ oe	B1 [1]
(b)	$U_1 = \left(\frac{25}{4}\right)\left(\frac{3}{5}\right)^1 = \frac{15}{4}$ oe	B1
	$S_n = \left(\frac{15}{4}\right)\left(\frac{3}{5}\right)^0 + \left(\frac{15}{4}\right)\left(\frac{3}{5}\right)^1 + \ldots\ldots + \left(\frac{15}{4}\right)\left(\frac{3}{5}\right)^{n-1}$	M1
	$\Rightarrow S_n = \sum_{r=1}^{n}\left(\frac{15}{4}\right)\left(\frac{3}{5}\right)^{r-1}$ oe	A1 [3]
	ALT 1	
	$U_1 = \left(\frac{25}{4}\right)\left(\frac{3}{5}\right)^1 = \frac{15}{4}$	[B1
	$r = \frac{3}{5} \qquad U_n = \frac{15}{4} \times \left(\frac{3}{5}\right)^{n-1}$	M1
	$S_n = \sum_{r=1}^{n} \frac{15}{4} \times \left(\frac{3}{5}\right)^{r-1}$	A1]
	ALT 2	
	$S_n = \sum_{r=1}^{n} \frac{25}{4}\left(\frac{3}{5}\right)^r$	[B1
	$= \sum_{r=1}^{n} \frac{25}{4} \times \frac{3}{5} \times \left(\frac{3}{5}\right)^{r-1}$	M1
	$= \sum_{r=1}^{n} \frac{15}{4} \times \left(\frac{3}{5}\right)^{r-1}$	A1]
(c)	$S_\infty = \frac{\frac{15}{4}}{1-\frac{3}{5}} = \frac{75}{8}$ or $9.375 \qquad S_n = \frac{\frac{15}{4}\left(1-\left[\frac{3}{5}\right]^n\right)}{1-\frac{3}{5}}$	M1M1
	$9.375 - \frac{\frac{15}{4}\left(1-\left[\frac{3}{5}\right]^n\right)}{1-\frac{3}{5}} < 0.045 \Rightarrow \frac{\frac{15}{4}\left(1-\left[\frac{3}{5}\right]^n\right)}{1-\frac{3}{5}} > 9.33$	
	$\frac{\frac{15}{4}\left(1-\left[\frac{3}{5}\right]^n\right)}{1-\frac{3}{5}} > 9.33 \Rightarrow 1-\left[\frac{3}{5}\right]^n > 0.9952 \Rightarrow \left[\frac{3}{5}\right]^n < \frac{3}{625}$	dM1
	$n \lg\left[\frac{3}{5}\right] < \lg\frac{3}{625} \Rightarrow n > 10.45$	ddM1A1
	$n = 11$	A1 [6]
		Total 10 marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 13 of 17 · Question 8 part a · 1 mark

2023 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
		Mark (a) and (b) together
(a)	B1	For the correct value only. Decimal value is 0.486.
(b)	B1	Finds the value of the first term. Must be identified as U_1 / first term.
	M1	Forms the sum of the series with at least the first term and the nth term.
	A1	For the correct expression.
	ALT 1	
	B1	Finds the value of the first term. Must be identified as U_1 / first term.
	M1	For finding r and writing $U_n = \frac{15}{4} \times \left(\frac{3}{5}\right)^{n-1}$
	A1	For the correct expression.
	ALT 2	
	B1	For correct summation statement using U_n from the question
	M1	For showing process to change index from r to $r-1$ and find the value of $\frac{A}{B}$ Minimum working is $\frac{25}{4} \times \left(\frac{3}{5}\right)^{r} = \frac{25}{4} \times \frac{3}{5} \times \left(\frac{3}{5}\right)^{r-1}$. May be with n as index.
	A1	For the correct expression.
(c)	Trial and improvement approaches for (c) should be sent to review.	
	M1	Correct method to find the sum to infinity using the correct formula with their $a = \frac{15}{4}$ and $r = \frac{3}{5}$. Their a should be $\frac{A}{B}$ if found in (b). Condone if working with $a = \frac{25}{4}$, r must be correct.
	M1	Using the correct formula, forms the sum to n terms using their $a = \frac{15}{4}$ and $r = \frac{3}{5}$. Their a should be $\frac{A}{B}$ if found in (b). Condone if working with $a = \frac{25}{4}$, r must be correct.
	dM1	Forms the inequality in terms of n with their S_∞ and S_n and simplifies to $\left[\frac{3}{5}\right]^n < '\frac{3}{625}'$ Must have dealt with negative. Allow for working with equation if inequality sign reinstated later. Reinstatement can be implied by rounding up to the next integer value once n found. Dep M1M1
	ddM1	Uses logarithms correctly to attempt to find the value of n They must reverse the inequality for this mark. Allow for working with equation if inequality sign reinstated later. Reinstatement can be implied by rounding up to the next integer value once n found. Dep previous M mark. Allow use of log base $\frac{3}{5}$.
	A1	For correct n awrt 10.5 Can be implied by $n = 11$ if decimal not seen.
	A1	For $n = 11$

Type MS · Finding the first term and common ratio of a geometric series

Answer 13 of 17 · Question 8 part a · 1 mark

2023 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 14 of 17 · Question 6 part a · 2 marks

2025 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

6(i)(a)	$\left[\frac{ar}{a}=r=\right]\frac{\sqrt{3}}{3+\sqrt{3}}\times\frac{3-\sqrt{3}}{3-\sqrt{3}}$ oe or $\frac{\sqrt{3}\left(3-\sqrt{3}\right)}{3^2-\left(\sqrt{3}\right)^2}$ $=\frac{3\sqrt{3}-3}{6}$ oe $\Rightarrow\frac{\sqrt{3}-1}{2}$ or $\frac{-1+\sqrt{3}}{2}$	M1 A1 [2]
(b) (i)	$\frac{\sqrt{3}-1}{2}\left(=0.366\right)<1$ or $\lvert r\rvert<1$ so the series is convergent	B1ft [1]
(ii)	$S=\frac{8\left(1-0.6^n\right)}{1-0.6}$, $T=\frac{8}{1-0.6}=\left[20\right]$ $\left["20"-"20"\left(1-0.6^n\right)<0.12\right]\Rightarrow 0.6^n<0.006$ oe eg $-0.6^n>-\frac{3}{500}$ $n\log_{10}0.6<\log_{10}0.006$ oe eg $n\log_{0.006}0.6<\log_{0.006}0.006$ $n>\frac{\log_{10}0.006}{\log_{10}0.6}$ $\left(n>10.015\right)$ $n=11$	M1, M1 M1 dM1 A1 [5]
		Total 8 marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 14 of 17 · Question 6 part a · 2 marks

2025 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For dividing the second term by the first term and for a correct method **stated** to rationalise the denominator. A correct answer may not imply this mark as candidates are told to show working.
	A1	For $\frac{3\sqrt{3}-3}{6}$ oe **and** $\frac{\sqrt{3}-1}{2}$ or $\frac{-1+\sqrt{3}}{2}$ Accept $p = -1$ and $q = 2$ Minimum steps as shown in the mark scheme needed, with no errors or omissions. Additional steps shown must be checked and correct.
If students start completing 1st term divide 2nd ie $\frac{3+\sqrt{3}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{3\sqrt{3}+3}{3}$ they must progress to $\frac{3}{3\sqrt{3}+3}$ oe eg $\frac{1}{\sqrt{3}+1}$ before marks can be awarded.		
(b) (i)	B1ft	For a correct explanation referring to r or the value being < 1 or $\lvert r \rvert < 1$ This may ft any value of r from their part (a) where $-1 < r < 1$ Accept "r is between 1 and – 1"
(ii)	M1	For the correct expression for S or T using the given values. **May be implied/embedded within the solution.**
	M1	For the correct expression for S and T using the given values. **May be implied/embedded within the solution.**
	M1	For an inequality of the form $0.6^n < A \quad A > 0$ or $-0.6^n > -B \quad B > 0$ $0.6^n < 0.006$ oe eg $-0.6^n > -\frac{3}{500}$ will definitely imply first 2 M marks if one or both are not explicitly stated.
	dM1	For using logarithms correctly, including reversing the inequality to find a value for n Dependent on previous method mark
	A1	For the correct value of n **following correct work.** Candidates must use logarithms and we must see all previous method marks. If ($n = 11$) comes from incorrect logs work or incorrect reversal of this inequality, this will be A0.

Type MS · Finding the first term and common ratio of a geometric series

Answer 14 of 17 · Question 6 part a · 2 marks

2025 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 15 of 17 · Question 1 part whole · 6 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

1 (i)/(ii)	$ar^2 = a+9d = 48$ or $a+9d = 4ar$	B1
	$\frac{4ar}{ar^2} = 1$	M1
	$\frac{4}{r} = 1$	M1
	$r = 4 \quad a = 3 \quad d = 5$	A1 A1 A1
		[6]
	Alternative method $a_1 + 9d = t_1 r^2$ or $a_1 + 9d = 4t_1 r$	B1
	$r^2 = 4r$	M1
	$r = 4$	A1
	$t_1 r = 12 \quad$ So $t_1 = a_1 = 3$	A1
	$a_1 r + 9dr = 48r \Leftrightarrow 12 + 36d = 192 \Leftrightarrow 36d = 180$	M1
	$d = 5$	A1
		[6]

	Notes
B1	For either $ar^2 = a+9d = 48$ or $a+9d = 4ar$ oe
M1	For solving simultaneously
M1	For simplifying to $\frac{4}{r} = 1$ oe
A1	$r = 4$
A1	$a = 3$
A1	$d = 5$
	Alternative
B1	For either $a_1 + 9d = t_1 r^2$ or $a_1 + 9d = 4t_1 r$ oe
M1	For solving simultaneously
A1	$r = 4$
A1	$a = 3$
M1	For $a_1 r + 9dr = 48r \Leftrightarrow 12 + 36d = 192 \Leftrightarrow 36d = 180$
A1	$d = 5$

Type MS · Finding the first term and common ratio of a geometric series

Answer 15 of 17 · Question 1 part whole · 6 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common ratio of a geometric series

Answer 16 of 17 · Question 7 part a · 5 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

7(a)	$\frac{(x+1)}{(x-3)}=\frac{(4x-2)}{(x+1)}$ or $(x+1)^2=(x-3)(4x-2)$	M1
	$(x+1)^2=(4x-2)(x-3)\Rightarrow 3x^2-16x+5(=0)$	A1
	$(x-5)(3x-1)=0\Rightarrow x=5,\ \frac{1}{3}$	M1A1A1 (5)
(b)	$x=\frac{1}{3}\Rightarrow r=\frac{x+1}{x-3}=\frac{\frac{4}{3}}{\frac{-8}{3}}$	M1
	$r=-\frac{1}{2}$ ∴ convergent as $-1<r<1$ *	A1cso (2)
(c)	$S=\frac{a}{1-r}=\frac{-\frac{8}{3}}{1+\frac{1}{2}}=-\frac{16}{9}$ oe	M1A1 (2)
(d)	$\frac{S}{S_n}=\frac{a}{1-r}\times\frac{1-r}{a(1-r^n)}=\frac{1}{(1-r^n)}=\frac{256}{255}$	M1
	$255=256(1-r^n)$	
	$256r^n=1$ $\left(-\frac{1}{2}\right)^n=\frac{1}{256}$ oe $n=8$	dM1A1 (3) [12]

(a)M1	Form an equation using the given information about the terms
A1	Simplify their equation to a correct 3TQ, terms in any order Condone missing = 0
M1	Attempt to solve their 3TQ by any valid method. Must reach x = ... (at least one root) Calculator solutions: *Both* roots correct from a correct equation scores M1A1A1 Incorrect equation or incorrect roots scores M0A0A0
A1	One correct value of x
A1	Both correct values of x
(b)	
M1	Use either of their values of x, provided it is < 1, to find the corresponding value of r. No need to simplify
A1cso	Correct value of r **and** the conclusion including the reason
(c)	
M1	Use their value of r (not x) provided $-1<r<1$ (as found in (b) or here) and the formula for the sum to infinity to obtain a value for S
A1	Correct value
(d)	
M1	Obtain an equation in r and n. May use the formulae to cancel a or may sub values of a and r in the formulae for the LHS Must equate to 256/255 Value of r not needed for this mark so allow any value used.
dM1	Solve their equation of the form $r^n=...$ where $-1<r<1$ (r not x). May use trial and improvement or logs. This mark can be given if the equation and value of r are incorrect. Evidence of method needed if final answer is incorrect. If logs used condone log $(-1/2)^n$
A1	Correct value from correct working

Type MS · Finding the first term and common ratio of a geometric series

Answer 16 of 17 · Question 7 part a · 5 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question Number	Scheme	Marks
	5	

	M1	$\left(6\sqrt{2}\right)$ or AC $\left(12\sqrt{2}\right)$
	M1	For using the correct trigonometry (tan ratio): $h = '6\sqrt{2}' \tan 30^\circ = \left[2\sqrt{6}\right]$ or equivalent. Eg $\tan 30 = \dfrac{h}{6\sqrt{2}} \Rightarrow h = ...$
	A1	For $2\sqrt{6}$ oe
(b)	M1	For correct expression for the midpoint of AD to F $\left(6 - \dfrac{12}{1+4}\right)$ or 3.6 oe seen
	M1	For the correct use of Pythagoras' to find OF $\dfrac{6\sqrt{34}}{5}$ oe [6.997..] ft their 3.6
	ALT	
	M1	Finds the length OF using cosine rule. $OF = \sqrt{2.4^2 + \left(6\sqrt{2}\right)^2 - 2\times 2.4\times 6\sqrt{2}\times\cos 45} = \left[\dfrac{6\sqrt{34}}{5}\right]$ OR $OF = \sqrt{9.6^2 + \left(6\sqrt{2}\right)^2 - 2\times 9.6\times 6\sqrt{2}\times\cos 45} = \left[\dfrac{6\sqrt{34}}{5}\right]$ OR $\cos 45^\circ = \dfrac{9.6^2 + \left(6\sqrt{2}\right)^2 - OF^2}{2\times 9.6\times 6\sqrt{2}} \Rightarrow OF = ...$ oe [6.997..]
	M1	For the correct evaluation of their cosine rule. $\left(\dfrac{6\sqrt{34}}{5}\text{ oe}\right)$ Accept awrt 7.00 [6.997142…]
	dM1	For the correct use of $\tan\theta = \dfrac{'EO'}{'OF'}$ This mark is dependent on both previous M marks. They must have a valid method to find OF for the award of this mark.
	A1	For 35(°) or better (Calculator value is 34.997…°.)
	Note: There are other methods – if unsure, send to Review.	

Type MS · Finding the first term and common ratio of a geometric series

Answer 17 of 17 · Question 6 part a · 5 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks
6 a	$r=\frac{q(2p-3)}{q(2p+3)}=\frac{q(2p+3)}{q(4p+1)}$	M1
	$(4p+1)(2p-3)=(2p+3)^2$	dM1
	$2p^2-11p-6=0$	A1
	$(2p+1)(p-6)=0$	M1
	$p=-\frac{1}{2}$ or $p=6$	A1 (5)
b	When $p=6$ $r=\frac{3}{5}$ and $U_1=q\left(4\times'6'+1\right)=\left(25q\right)$	M1
	$S_\infty=\frac{25q}{\frac{2}{5}}=250\Rightarrow q=4$	dM1 A1 (3)
		Total 8 marks